

Summary of Spacecraft Design Problems

Introduction

Behind the Scenes (from a visitor to the Jet Propulsion Laboratory)

In a few minutes, you will hear a summary of the problems that engineers faced when they started designing a spacecraft to go to Saturn. I visited the Jet Propulsion Laboratory in Pasadena and talked to the engineers who built the spacecraft that is going to Saturn. It was cool to visit there! You get to see mission control where they track the progress of lots of different spacecraft, and you can see a room that looks like Mars where they are testing rovers to see how they might work on Mars. The engineers are very excited about their spacecraft. I took 26 pages of notes! Then I needed to decide what to write about, because I don't think you want to hear all 26 pages, do you? I decided to list the most common problems that engineers face when they are building a spacecraft. There are millions of other problems that engineers faced too, but some of them you and I would have no idea how to solve unless we got a degree in engineering. I hope you enjoy hearing this summary.

Designing a Spacecraft

Today you are going to need to think like an engineer. Many of the people who work at NASA are engineers. Engineers at NASA design instruments to help scientists discover things.

Here is your design problem for today. How can you design a spacecraft that will make it all the way to Saturn — 800 million miles away from Earth? To design your spacecraft, you and your partner are going to work together and make decisions about the following things.

- How will you keep the spacecraft from freezing in the icy cold temperature of deep space?
- How will you control and keep in touch with your spacecraft?
- How will you keep the spacecraft safe if it is hit by a space particle (a micrometeoroid, dust particle, or ring particle)?
- What do you want the spacecraft to try to find out at Saturn?
- Do you think you should send humans on the spacecraft? If you think you should send humans, how will they survive?
- What are some things that could go wrong with the spacecraft on the journey to Saturn?
- What will you name your spacecraft?
- Why did you choose that name?

Once you think of possible solutions to these design problems, you and your partner are going to draw a sketch of your Saturn spacecraft!

